

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITUEA2-44

Group: 1

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature 	Date 05/11/2025
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Lecturer's Comments:

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Marks Awarded:	%
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Signature	Date
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Section A

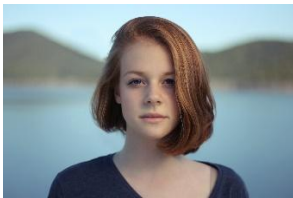
Question 1

1.1) User groups:

- Users who want a healthier lifestyle and want to consistently maintain a healthy diet.
- Users who have specific dietary requirements due to personal reasons such as being a vegan or having religious restrictions.

A detailed persona refers to a description of a specific user that indicates the user's needs. It is used to guide the development direction of the product (Jolanda et al., 2025).

A detailed persona:



(Campbell, C. 2015)

Robin Lee is a 24-year-old university student that has classes every day where they have very minimal free time. The restriction on time prevents her from researching and experimenting with new dishes, which is made difficult by the fact that she is a vegetarian and has a peanut allergy. Her main goal is to be made aware of and create new dishes within her strict scheduling where these dishes also meet her dietary requirements. In addition, it is also valuable for Robin to cognitively offload her memory when planning meals as she has other aspects of her life that requires her memory.

An empathy map refers to visual tool used in usability engineering to determine what designers understand from specific users and their needs, which can assist with decision making. An empathy map is made up of four quadrants including what the user says, thinks, does and feel about the product (Huang et al., 2025).

Therefore, an appropriate empathy map for the Robin Lee user is:

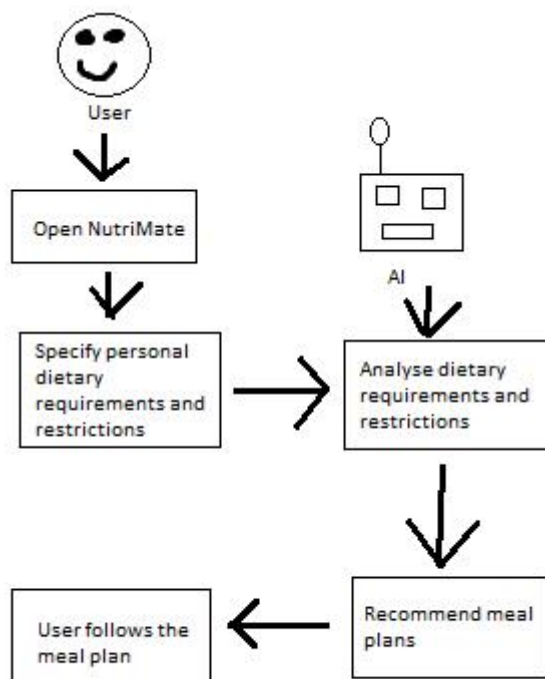
What they say	What they do
"NutriMate helps me to find meals that I didn't know existed."	The user uses NutriMate to remember their meal plan.
What they think	What they feel
Will NutriMate provide me meals that respect my dietary needs? Is NutriMate trustworthy by ensuring that my privacy is protected?	They feel worried, because they don't feel healthier after following the meal plans. They feel worried that the AI might not follow their requirements.

1.2) Interaction design refers to creating products that can greatly enhance how individuals are able to interact with various aspects of their lives. It covers a large scope of different technologies and fields that can have an impact on the usability of the product and the user experience.

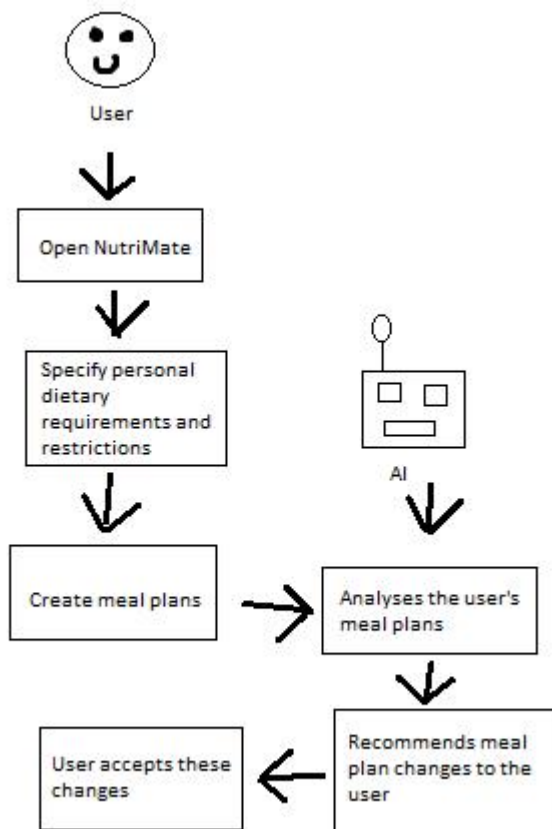
Interaction design is relevant to NutriMate as an interactive system, because usability is an important factor in a system’s success where a user interface that is considered unusable will lead to an unsuccessful interactive system (Ammar et al., 2016). This application requires the unique dietary requirements from the user in order for it’s AI to plan meals that are best suited for the user based on any restrictions and requirements. If the user can’t accurately indicate their requirements and restrictions to NutriMate, the meals that the AI in NutriMate suggests will be unsuitable for the user. Therefore, it can lead to negative consequences such as the user using another application that is more reliable than NutriMate or experiencing health issues due to the planned meals.

1.3)

Conceptual model 1:



Conceptual model 2:



An interface metaphor refers to designing interface concepts and actions in a manner that makes it easier for users to understand its function by relating it to other concepts from the user's knowledge (Wells et al., 2005). An interface metaphor that can be used for NutriMate is a "Grocery List" button that shows the ingredients of recipes provided by NutriMate. This metaphor interface is effective for its intended use, because a grocery list is a list of items, that can include dish ingredients, that someone uses to remember what they should buy while shopping. It enables the user to use it as an actual grocery list when buying the ingredients for the prescribed dishes.

1.4) The evolution of interaction design between simple graphical interfaces to modern technologies occurs due to the advancements of technology and the introduction of new tools that can impact how a product feels. New technology and advancements increases the complexity of interaction design as designers must be able to master and utilise these new tools to ensure that product designs are still relevant to the current time (*Discover Artificial Intelligence, 2025*). The increased complexity can make interaction design more effective than before, because designers are able to implement new components that can greatly improve the user experience.

The current trends for mobile/AI technology relates to the evolution of interaction design as it provides new technological tools that can be used by designers to improve the user experience. The current trend for artificial intelligence (AI) technology is implementing machine learning where a machine has the ability to make decisions and predictions by learning from a dataset (Najeeb et al., 2025). These AI technologies can be accessed on mobile devices, which can enhance

interaction design as designers can utilise these AI components in their products to improve the user experience.

For NutriMate, implementing AI components can assist the user with meal planning as the AI can suggest and create meal plans according to the user's dietary restrictions. This could lead to an improvement in user experience as the user can be introduced to a meal plan that is more suitable for them due to their conditions.

Question 2

2.1)

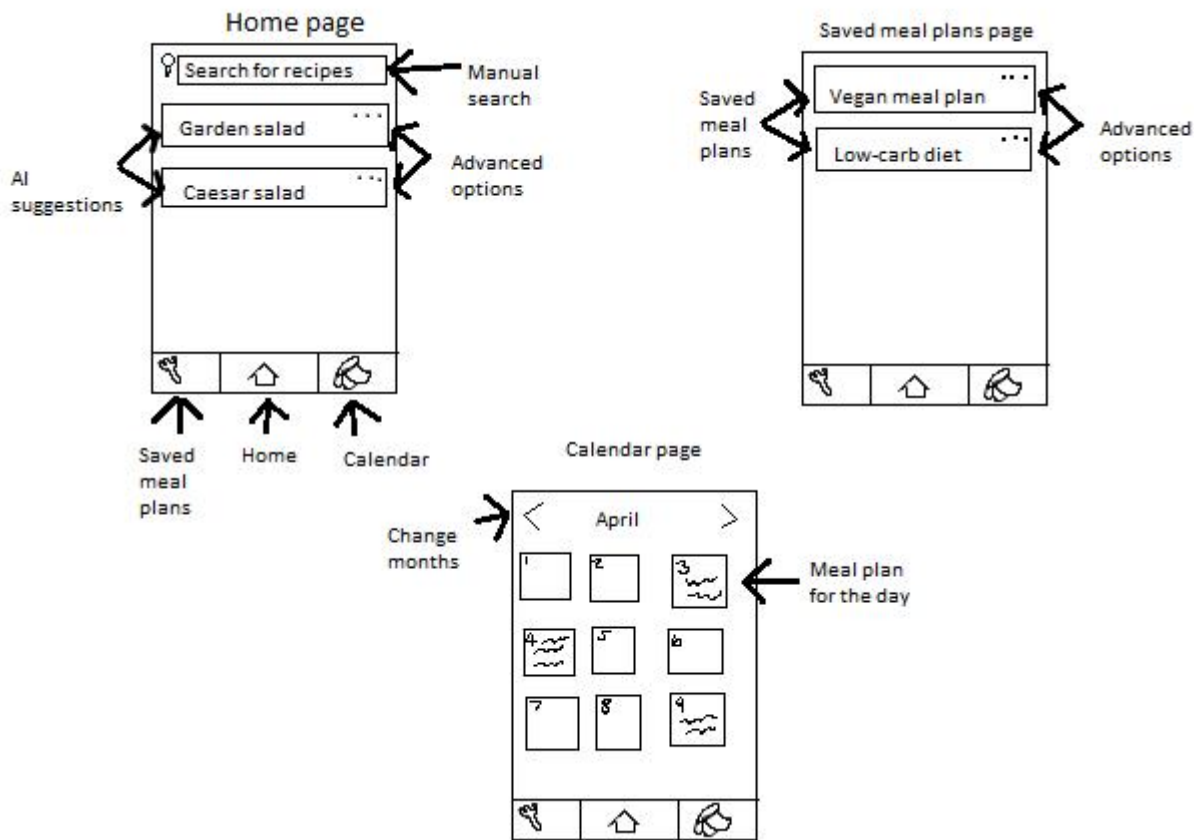


Figure 1: Low-fidelity prototype

According to figure 1, the user will be able to search for specific recipes with the assistance of the application's AI capabilities that will recommend similar dishes. The advanced options for both the recipes and meal plans provides additional options that the user can perform such as the ability to share these recipes and meal plans to other individuals. The user can also create meal plans consisting of recipes chosen by the user that will be scheduled on the application's calendar page.

The prototype in figure 1 indicates various design decisions that can assist in reducing the cognitive load on the user. An example can be the "Saved meal plans" page that stores all the user's meal plans as well as a "Calendar" page that indicates what meal plan should be followed each day of the month, which can assist the user with cognitive offloading their memory onto the application. In addition, the user can offload their perception as the application offers some visual cues that can make it easy for the user to understand what a feature does such as the fork icon on the bottom left of the application that represents the "Saved meal plans" page, since it's an eating utensil. Moreover, the magnifying glass on the left-hand side of the search bar provides further indication of the search bars purpose, promoting consistency with other applications like internet browsers that also utilise a magnifying glass for their search bar. Furthermore, problem-solving can also be offloaded by the user by having the application's AI capabilities recommend recipes based on the user's current meal plans and restrictions specified by the user such as a peanut allergy, which can result in the user spending less time in determining recipes that are most suitable for them when creating new or adding to existing meal plans.

2.2) I would recommend the recipe and saved meal plan buttons to be 4 x 2 cm with 1 cm of space between each of these buttons. Having each individual button have a length of 4 cm and a height of 2 cm makes the text on each button readable for most users. Additionally, the space between these buttons can prevent the user from pressing the incorrect button.

For the three application page buttons at the bottom of the app as shown in figure 2, a 2.5 x 2.5 cm size is recommended. This size can ensure that all three buttons are big enough to display it's associated icon that can be recognised by the user easily. Moreover, the size of each button is big enough to prevent the user from pressing the incorrect button while also being able to reasonably fit within the horizontal space of a device.

2.3) According to figure 1, the user can create a weekly menu by finding recipes in the "Home" page that is suggested by the AI capabilities of the application or by manually searching for specific dishes. After finding appropriate recipes, the user can then add them to a meal plan with the advanced options provided by the application, where this new meal plan will be stored in the "Saved meal plans" page. Furthermore, the user can use the advanced options of the meal plan to specify that the user wants the meal plan to be weekly, which can be viewed on the "Calendar" page. The application can then provide menu suggestions to the user based on the day of the month and the meal plan associated with it.

Question 3

3.1)

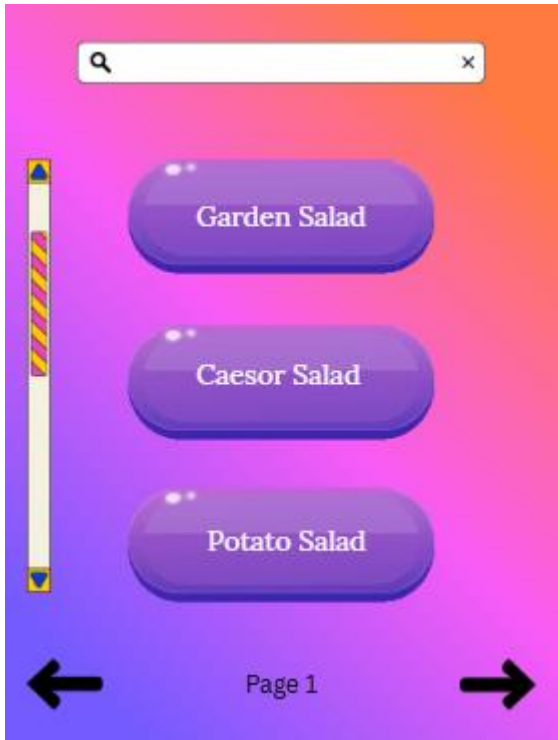


Figure 2: Clickable prototype

Various aspects of the clickable prototype in figure 2 are designed to mitigate or minimise the issues discovered in the previous usability study including an overwhelmingly high-quantity of results and small tap targets.

The prototype in figure 2 divides the recipes into separate pages that the user can only access by pressing the arrows at the bottom to swap between pages. This feature is designed to prevent users from feeling overwhelmed by a large number of recipes while still having access to all the recipes and reduce the frequency of back-and-forth movements. Moreover, the recipe buttons in figure 2 have been scaled to a larger size and are more spread out from each other than the previous prototype to prevent misinputs from users.

3.2)

The test used to determine if the new prototype in figure 2 meets the success criteria includes the following tasks that the participants must perform:

- Task 1: Does the user understand the new page selection function?
- Task 2: Does the user find the quantity of recipes on a single page overwhelming?
- Task 3: Does the user continue to perform misinputs?

The success criteria includes that:

- Task 1: +80% of users understood the feature's function and understood it within 30 seconds.
- Task 2: +70% of users were not overwhelmed and were able to find their desired recipe within one minute.

- Task 3: +70% of users did not experience any misinput issues.

Each participant will be asked for their consent to provide feedback on the prototype from figure 2. Each participant that gives their consent will be given this form where they will be able to report their experience with the new prototype, enabling the success metrics to be measured and any unknown issues to be discussed.

Participant name:	
Did you understand the new page selection function within a short period of time?	
Did you find the quantity of recipes on a single page overwhelming?	
Did you perform any misinputs like pressing the wrong button?	
Additional feedback	

3.3)

Feedback received from the participants include:

- Participant 1 (Brother)
 - Task 1: They had some difficulty at first with understanding that they could press the arrows to swap between pages. However, they were able to swap pages within the 30 second metric.
 - Task 2: They commented that they found the recipe list easier to navigate than the previous prototype and were able to find their desired recipe within 20 seconds.
 - Task 3: They did not perform any misinputs.
- Participant 2 (Sister)
 - Task 1: They understood the page selection feature immediately; however, they suggested that the arrow buttons could be improved visually to look more interactable than it is currently.
 - Task 2: They had no issues and found it within the time metric
 - Task 3: They did not perform any misinputs.
- Participant 3 (Father)

- Task 1: They did not understand the page selection feature until after the one-minute metric.
- Task 2: They felt less overwhelmed by the number of recipes in the current prototype than the previous prototype, however they suggested that the quantity of recipes should be lessened further. They were able to find their recipe within the time metric
- Task 3: They commented that they found the buttons easier to press than the previous prototype, however they did struggle with reading the recipe names on the buttons.
- Participant 4 (Grandmother)
 - Task 1: They had great difficulty with understanding the concept of dividing the number of recipes into separate pages and figured it out after one minute.
 - Task 2: They were still overwhelmed by the quantity of recipes and only found their recipe after the one-minute metric.
 - Task 3: They commented that they enjoyed the new enlarged recipe buttons and had no misinput issues with them.
- Participant 5 (Mother)
 - Task 1: They understood the page selection feature within the one-minute metric, but they did suggest that it should be designed in a manner that makes its use more apparent to users.
 - Task 2: They had no issues and found it within the time metric
 - Task 3: They did not perform any misinputs, however complained that the recipe buttons were too big and were not visually appealing.

According to the success metrics:

- 60% of the participants understood the feature's function and understood it within 30 seconds.
- 80% of participants were not overwhelmed and were able to find their desired recipe within one minute.
- 100% of participants did not experience any misinput issues.

Task 1 indicates a failure, while tasks 2 and 3 indicate a success. Therefore, the new prototype is an overall success where issues such as potential misinputs and an overwhelming quantity of recipes were mitigated. However, improvements should be made to further mitigate these issues such as improving the design of the page selection feature to make it's function more apparent to users than the current prototype, improving the readability of the text and further minimising the quantity of recipes on a page.

Question 4

4.1) Social interaction in this context refers to individuals interacting with each other in a multiuser environment, where each user can present themselves to others as well as receive feedback from other users in the environment (Xu et al., 2019). Social interaction that utilises technology as a medium can be accompanied by concepts such as collaborative tools that enables individuals to easily work with each other over the technology medium.

The community board feature that is proposed by NutriMate appears to utilise the concept of collaborative tools to enhance social interaction through the ability to share meal plans and tips. Sharing meal plans with other users can enable them to become inspired as they can be made aware of dishes that exist in different areas around the world and can then incorporate these new dishes to their meal plans. Moreover, the ability for users to exchange tips with each other enables users to educate and provide suggestions to each other on how they can improve upon their meal plans with the possibility for users to become friends over NutriMate.

Furthermore, the motivational badges that are rewarded to users can further promote social interaction between users by motivating other users to improve their own meal plans if the original user provides guidance. The badge can prove that the original user is experienced with consistent healthy eating, and their guidance can be more trustworthy for users who are not willing to listen to other user's advice. Moreover, some users can also showcase their motivational badges, because they are proud of their achievements that could result in other users following their example.

4.2) There are various issues that can occur on NutriMate's community board feature due to the ability for multinational groups to co-exist. Users from different areas in the world tend to struggle with fluency of other language, which can prevent them from understanding instructions provided by NutriMate. Furthermore, the language barrier also prevents meaningful discussions between other multinational groups that don't primarily use the same language as them (Qiaoyan et al., 2025). Additionally, the ability for users of NutriMate to interact with overlapping social identities from different multinational groups can lead to them experiencing discrimination from other users (Dzidzai et al., 2025). These users can utilise the social features of the community board, like the ability to exchange tips, to spread hate and discriminate against certain groups or to instigate conflicts to due cultural differences.

The language barrier between multinational groups can be minimised by expanding the AI capabilities of NutriMate to automatically translate recipes, tips and meal plans to the chosen language of the user. The translation might not always be completely accurate in its translation; however, it can still bridge the gap between multinational groups by enabling them to communicate and interact with each other.

Additionally, potential discrimination and conflicts that can occur in NutriMate can be minimised by introducing a report feature that users can use against problematic users. The report function can highlight users that make the community environment feel hostile to certain multinational groups and enables NutriMate to take appropriate action to solve the issue, such as banning the user from the application or restricting their access to the community features.

4.3) Persuasive techniques refer to techniques that are utilised to encourage users to perform specific actions (Pit et al., 2022). Persuasive techniques can be used for NutriMate's community

board feature to promote social interaction such as additional badges based on interacting with the community and a community recommendation mechanism.

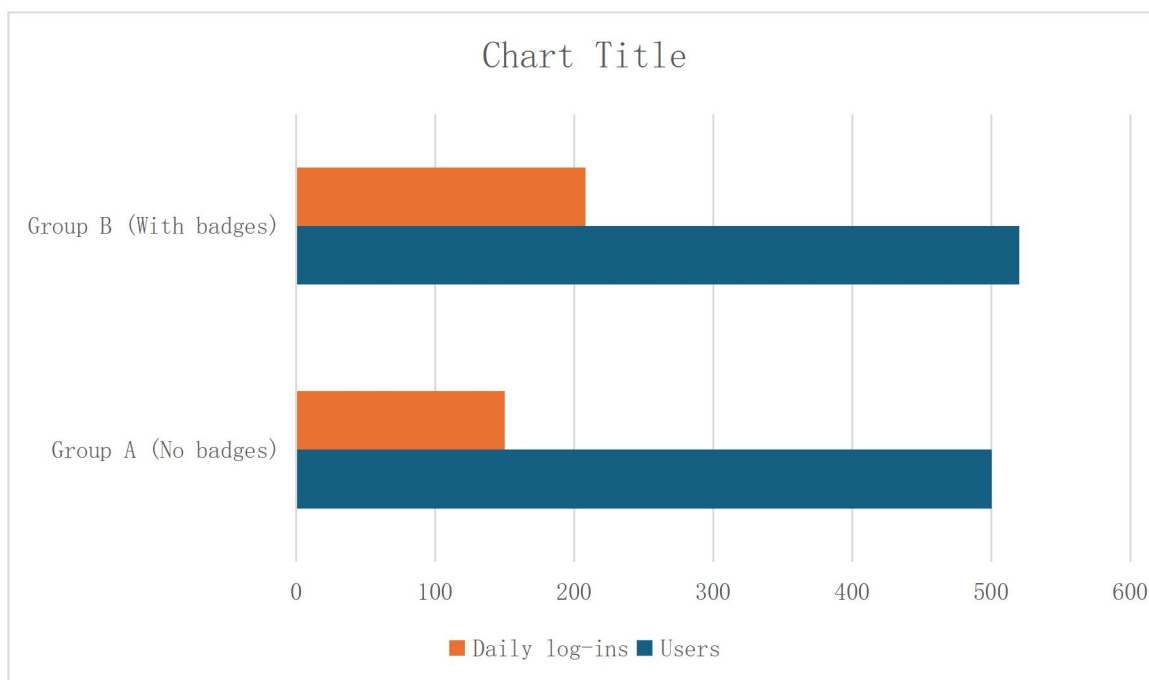
Additional badges that requires the user to interact with the community like providing tips to other users and sharing meal plans can persuade users, similar to the existing motivational badges, to be a part of NutriMate’s community. Persuading these users to interact with NutriMate’s community can enable them to be exposed to tips and meal plans offered by other users that can be incorporated it into their meal plans. Therefore, users can become inspired after viewing through a new perspective offered by other users, while also offering their perspective to other users.

Moreover, a recommendation feature that can recommend specific communities within NutriMate to a user, due to what meal plans they are trying to follow, can persuade the user to interact with users that are similar to them. These specific communities can enable the user to interact with and receive information that is tailored to their specific considerations, because the other users in the community have the same or very similar considerations. Therefore, the user is likely to feel a sense of belonging by interacting and potentially becoming part of a community filled with like-minded individuals.

4.4)

Conversion rate:

- Group A (No badges) = $(150 / 500) * 100 = 30\%$
- Group B (With badges) = $(208 / 520) * 100 = 40\%$



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